EXP 6

CODE:

```
# Tic Tac Toe Game (2 players)


def print_board(board):
    for row in board:
        print(" | ".join(row))
        print("-" * 5)


def check_winner(board, player):
    # Check rows, columns, diagonals
    for i in range(3):
        if all(board[i][j] == player for j in range(3)): return True
        if all(board[j][i] == player for j in range(3)): return True
    if all(board[i][i] == player for i in range(3)): return True
    if all(board[i][2-i] == player for i in range(3)): return True
    return False


def tic_tac_toe():
    board = [[" "]*3 for _ in range(3)]
    players = ["X","O"]
    turn = 0


    for move in range(9):
        print_board(board)
        player = players[turn]
        print(f"Player {player}'s turn")
```

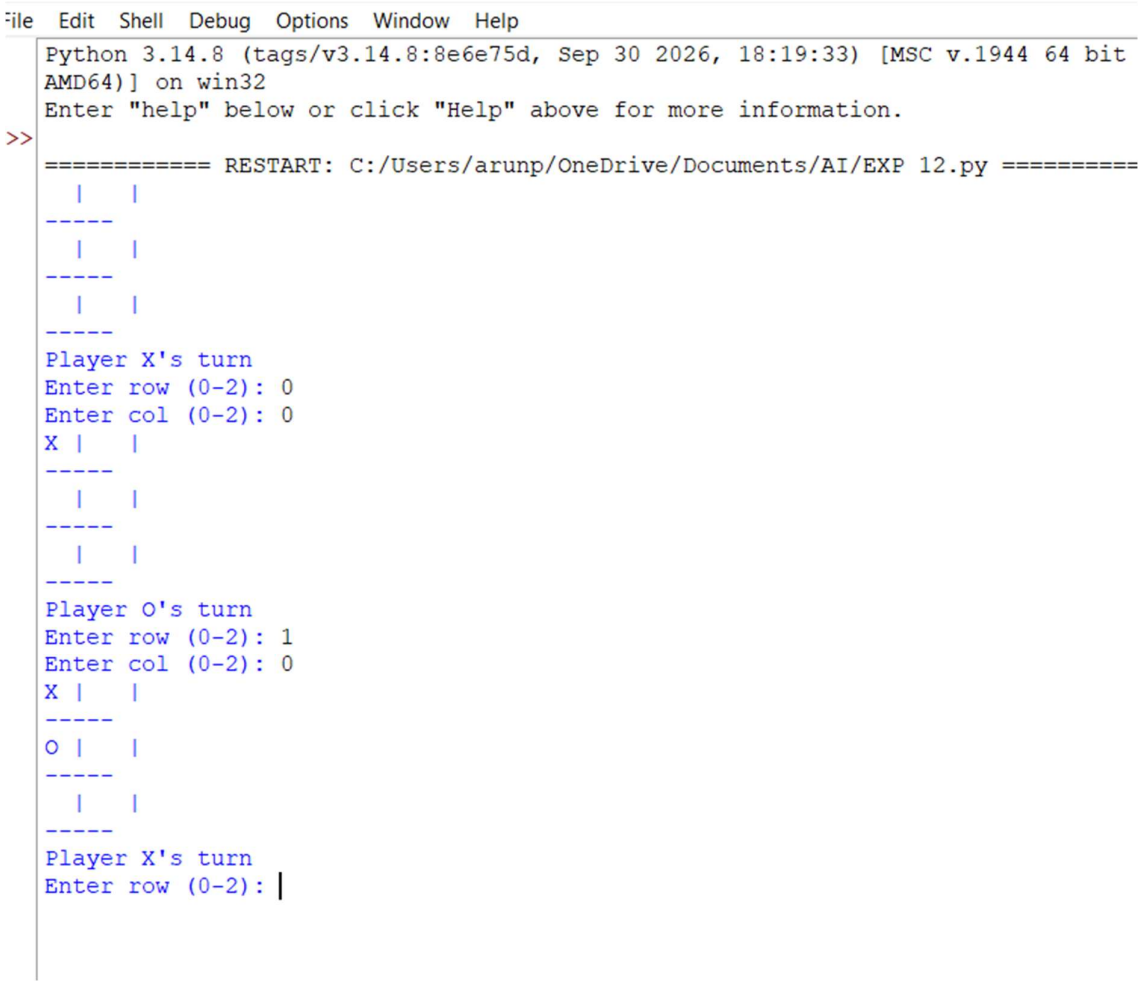
```
        # Input row and column
        r = int(input("Enter row (0-2): "))
        c = int(input("Enter col (0-2): "))

        if board[r][c] == " ":
            board[r][c] = player
            if check_winner(board, player):
                print_board(board)
                print(f"Player {player} wins!")
                return
            turn = 1 - turn
        else:
            print("Cell occupied! Try again.")
            move -= 1  # retry same move

    print_board(board)
    print("It's a draw!")

# Run the game
tic_tac_toe()
```

OUTPUT:


File Edit Shell Debug Options Window Help
Python 3.14.8 (tags/v3.14.8:8e6e75d, Sep 30 2026, 18:19:33) [MSC v.1944 64 bit AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>
========== RESTART: C:/Users/arunp/OneDrive/Documents/AI/EXP 12.py ==========
| |

| |

| |

Player X's turn
Enter row (0-2): 0
Enter col (0-2): 0
X | |

| |

| |

Player O's turn
Enter row (0-2): 1
Enter col (0-2): 0
X | |

O | |

| |

Player X's turn
Enter row (0-2):